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# Enhanced Thermal Boundary Conductance across GaN/SiC Interfaces with AlN Transition Layers

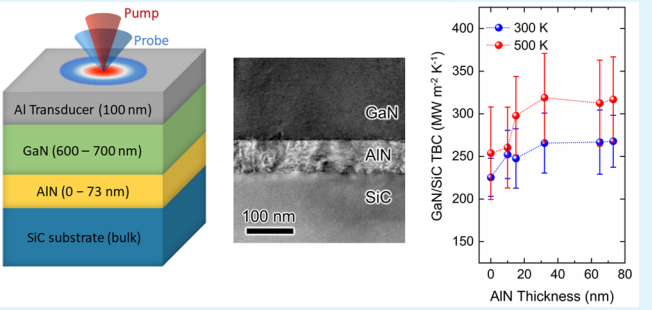
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**ABSTRACT:** Heat dissipation plays a crucial role in the performance and reliability of high-power GaN-based electronics. While AlN transition layers are commonly employed in the heteroepitaxial growth of GaN-on-SiC substrates, concerns have been raised about their impact on thermal transport across GaN/SiC interfaces. In this study, we present experimental measurements of the thermal boundary conductance (TBC) across GaN/SiC interfaces with varying thicknesses of the AlN transition layer (ranging from 0 to 73 nm) at different temperatures. Our findings reveal that the addition of an AlN transition layer leads to a notable increase in the TBC of the GaN/SiC interface, particularly at elevated temperatures. Structural characterization techniques are employed to understand the influence of the AlN transition layer on the crystalline quality of the GaN layer and its potential effects on interfacial thermal transport. To gain further insights into the trend of TBC, we conduct molecular dynamics simulations using high-fidelity deep learning-based interatomic potentials, which reproduce the experimentally observed enhancement in TBC even for atomically perfect interfaces. These results suggest that the enhanced TBC facilitated by the AlN intermediate layer could result from a combination of improved crystalline quality at the interface and the "phonon bridge" effect provided by AlN that enhances the overlap between the vibrational spectra of GaN and SiC.



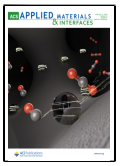

**KEYWORDS:** *thermal boundary conductance, GaN-on-SiC, time-domain thermoreflectance, phonon bridge, aluminum nitride*

## 1. INTRODUCTION

Gallium nitride (GaN), as a wide-bandgap semiconductor material, has garnered significant attention for its exceptional properties, including high breakdown voltage and carrier mobility. High electron mobility transistors (HEMTs) based on GaN are considered as key devices for high-power and high-frequency electronic applications, such as power amplifiers for satellite base stations and radar sensors.[1,2] Nonetheless, the performance of GaN-HEMTs is hindered by self-heating, which elevates the channel temperature and severely compromises performance and reliability.[3] To enhance heat dissipation, silicon carbide (SiC) has been utilized as a substrate due to its high thermal conductivity (~350 W $m^{-1}$ $K^{-1}$)[4] and relatively close lattice constant to GaN. However, the GaN/SiC interface typically exhibits limited thermal boundary conductance (TBC),[5,6] preventing the full utilization of the high thermal conductivity substrate.[7] Consequently, numerous experimental and theoretical studies have been conducted to investigate and enhance thermal transport across GaN/SiC interfaces.[7−15]

In the heteroepitaxial growth of high-quality GaN, an aluminum nitride (AlN) transition layer with a nanometer thickness is commonly employed to mitigate the lattice mismatch between GaN and SiC.[16] Some experimental studies have reported that the AlN layer acts as a thermal barrier for heat conduction.[8,9,11] It is believed that defects within the AlN film and near the interfaces induce substantial phonon-defect scattering, significantly impeding the thermal transport. The addition of the AlN layer itself also presents a thermal resistance. Recent research has observed high TBC (~230 MW $m^{-2}$ $K^{-1}$) for GaN/SiC interfaces without a transition layer.[7,15] However, the AlN transition layer is essential for ensuring a high GaN quality for electronic performance. Thus, optimizing the quality of the GaN/AlN and AlN/SiC interfaces, as well as the microstructure of the AlN film, becomes crucial to increasing TBC[17,18] while maintaining the

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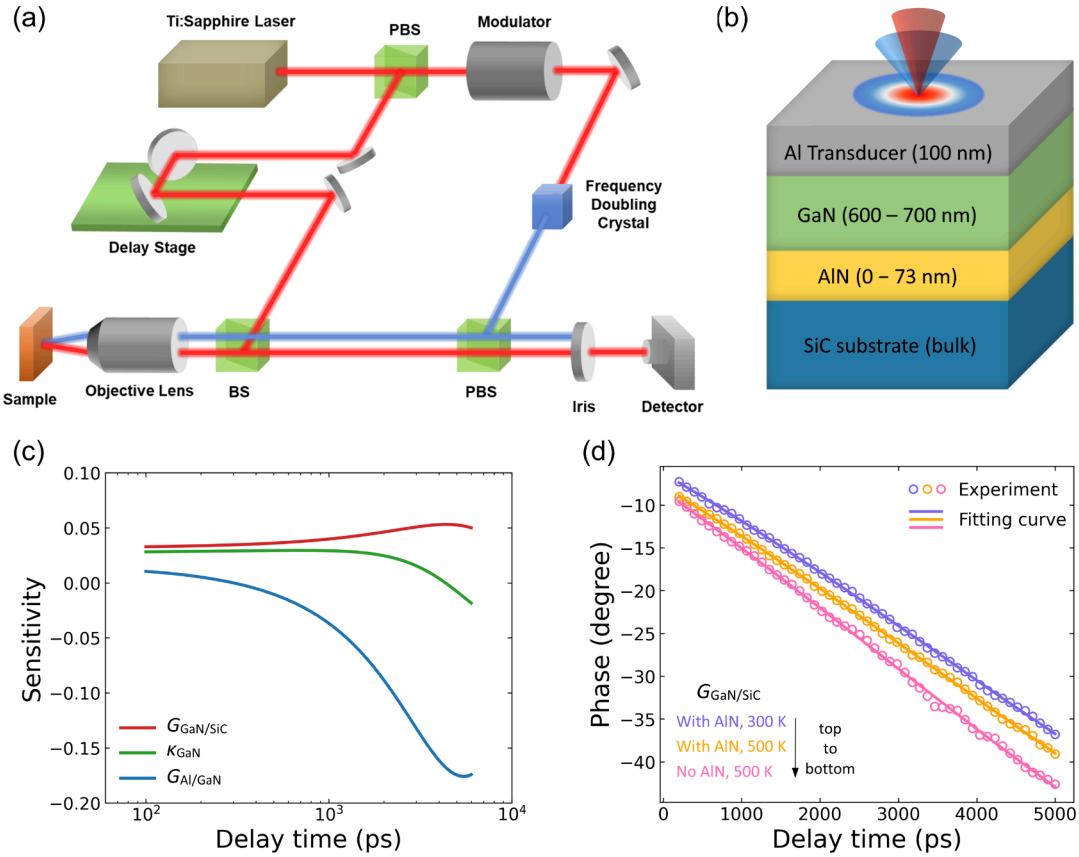



**Figure 1.** (a) Schematic of the TDTR pump–probe measurement. (b) Schematic of the sample geometry. (c) TDTR sensitivity of the phase signal to each fitting parameter: $G_{Al/GaN}$, $\kappa_{GaN}$, and $G_{GaN/SiC}$. (d) TDTR phase signals and best-fitted curves for sample V with a ~65 nm-thick AlN transition layer at 300 and 500 K, and for sample I at 500 K, which has no AlN.

high crystalline quality of GaN. Meanwhile, efforts have been made to remove the nucleation layer while preserving the high quality of the GaN layer using various pretreatment and growth techniques.[15,19,20] Mu et al.[15] employed a bonding method to directly bond high-quality GaN to SiC, achieving a high TBC interface of ~230 MW $m^{-2}$ $K^{-1}$. However, a direct comparison of TBCs for GaN/SiC interfaces with and without the AlN layer under identical growth conditions remains unexplored.

Intuitively, the insertion of an intermediate layer between two materials would be expected to increase thermal resistance, as observed in previous experiments.[21,22] However, recent computational and experimental studies have demonstrated that specifically engineered interfacial imperfections and interlayers can potentially enhance TBC across solid interfaces.[13,14,23−28] Atomistic simulations, in particular, have revealed that a thin and appropriately tailored intermediate layer between two solids can bridge their vibrational spectra and improve the interfacial thermal transport.[12,23,28,29] For the GaN/SiC interface, Lee and Luo[12] conducted molecular dynamics (MD) simulations using empirical Tersoff potentials[30] and observed an enhancement in TBC when the interface was coupled with an AlN layer, maximizing the overlap between the vibrational power spectra of GaN and SiC. This so-called "phonon bridge" effect has recently been experimentally studied by Giri et al.,[31] where confined solid-solution thin films between parent semiconducting materials did not necessarily increase thermal resistance. Thus, the question arises of whether such a phonon bridge effect can be observed for GaN/SiC interfaces with an AlN intermediate layer when directly comparing TBCs of interfaces with and without the AlN layer under the same growth conditions.

In this study, we experimentally demonstrate that the AlN transition layer can enhance thermal transport across GaN/SiC interfaces, attributable to the improved interface and crystalline quality as well as the aforementioned phonon bridge effect. The TBC enhancement is more pronounced at 500 K compared to that at 300 K. Specifically, we have prepared six samples with AlN transition layers of varying thicknesses (ranging from 0 to 73 nm) grown on SiC prior to the epitaxial growth of GaN. The effective TBC of GaN/SiC interfaces is measured for these samples utilizing time-domain thermoreflectance (TDTR, Figure 1a). Material characterization techniques, including high-resolution scanning transmission electron microscopy (STEM), energy-dispersive X-ray spectroscopy (EDS), and triple-axis X-ray diffraction (TAXRD), are employed to validate the sample quality and aid in the interpretation of the measured TBC results. Moreover, to reinforce our experimental findings, MD simulations are conducted using high-fidelity deep learning-based interatomic potentials, which further confirm the observed enhancement in TBC.

## 2. METHODS

We grew the GaN/AlN/SiC samples using low-pressure metal–organic chemical vapor deposition (LP-MOCVD). The epitaxial depositions of GaN were carried out on the Si face of 6H-SiC substrates (II–VI Inc.). Trimethylaluminum (TMA), trimethylgallium (TMG), and ammonia ($NH_3$) were used as the precursors for Al,

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Ga, and N, respectively. Prior to the growth, the SiC substrates were annealed at 1200 °C for 10 min in a $H_2$ ambient environment. This was followed by the growth of the AlN and GaN layers of the structure at 1050 °C and 150 Torr. For the six samples in this study, the GaN layer thicknesses are approximately uniform (600−700 nm) while the AlN buffer layer thickness varies from 0 to 73 nm. A sample with no AlN buffer layer was also grown to serve as a baseline for the investigation. For the growth of the AlN buffer layers, a pulsed epitaxy approach was used.[32] This enables the growth of high-quality layers at growth temperatures lower than what is needed for conventional epitaxy while still maintaining excellent crystalline quality. Moreover, it allows very precise thickness control that scales with the number of growth pulses. The AlN growth rate for this study was adjusted to 0.6 nm/pulse, which translates to 0.15 nm/s. The samples used in this work are denoted as samples I−VI, with increasing AlN thickness (sample I: 0 nm, II: 10 nm, III: 15 nm, IV: 32 nm, V: 65 nm, VI: 73 nm). It should be noted that an uncertainty of several nanometers exists in the determination of the AlN thickness. Cross-sectional STEM images of all six samples are provided in the Supporting Information (Figure S1), where we can see distinct boundaries between different layers.

We measured the effective TBCs of the GaN/SiC interfaces ($G_{GaN/SiC}$) using the TDTR technique (Figure 1a), which is an optical pump−probe technique widely employed for TBC measurements across various interface types.[15,27,33,34] Figure 1b illustrates the sample geometry used in our TDTR measurements. Prior to the measurement, a layer of Al (~100 nm) was deposited on each sample as an optothermal transducer using electron-beam evaporation. In a TDTR measurement, a modulated pump beam (400 nm) heats the sample surface while a delayed probe beam (800 nm) detects the temperature decay of the surface via thermoreflectance. The delay time is controlled by a mechanical delay stage, and the signals are collected by a photodetector with a lock-in amplifier. The measured signals are fitted with a heat transfer model to extract unknown thermal properties (e.g., $G_{GaN/SiC}$). More details of the experimental setup and data fitting can be found in refs 35−37 and the Supporting Information.

To infer unknown thermal properties, we utilize a three-layer model that includes the Al layer, the Al/GaN interface, the GaN layer, the GaN/SiC interface, and the SiC substrate in data fitting. Given the presence of relatively thin AlN transition layers (≤73 nm) and the imperative to maintain TDTR sensitivity to the buried interface, we have treated the AlN layer as an integral part of the GaN/SiC interface. This treatment effectively incorporates the AlN thermal resistance with the GaN/AlN and AlN/SiC boundary resistances. Importantly, this approach facilitates a consistent and meaningful comparison of TBCs for GaN/SiC interfaces, both with and without the AlN layer. A similar treatment was adopted in a previous experimental study to combine AlN-induced resistances into an effective AlN resistance in GaN/AlN/SiC composites.[11] The TBC of Al/GaN ($G_{Al/GaN}$), the thermal conductivity of GaN ($\kappa_{GaN}$), and $G_{GaN/SiC}$ are designated as fitting parameters. Other parameters, such as the thermal conductivity of SiC and Al, and the heat capacities of Al, GaN, and SiC, are fixed during the data fitting (see the Supporting Information).[38−41] This fitting approach has been employed and validated in prior TDTR measurements, particularly in scenarios involving buried interfaces with more than two unknown parameters.[11,15,42] The viability of concurrently fitting three unknown parameters can be assessed through sensitivity analysis. The sensitivities of TDTR measurement to the fitting parameters (at 300 K and 3.0 MHz pump beam modulation frequency) are shown in Figure 1c. Sensitivity to a parameter $x$ is calculated as $\partial\phi/\partial \ln x$, where $\phi$ represents the phase calculated from the heat transfer model of TDTR. As observed, the sensitivity to each parameter follows a different trend, which is favorable for extracting these unknowns with least-square minimization. For example, as depicted in Figure 1d, the experimental signals for sample V at both 300 and 500 K, along with the signals for sample I at 500 K, are well fitted to the thermal model curves.

To complement the experimental analysis, we also calculated the TBCs through MD simulations using high-fidelity neural network potentials (NNP) specifically designed for GaN/SiC and GaN/AlN/SiC interfaces. The training and testing data sets for the NNP development are generated by *ab initio* MD (AIMD) using the Vienna Ab initio Simulation Package (VASP)[43] under the local density approximation (LDA). The NNPs are trained using the scheme developed by Zhang et al.,[44] which has been proved capable of accurately simulating solid interfaces, such as Si/Ge interfaces.[45] Two NNPs are developed for GaN/SiC and GaN/AlN/SiC, respectively, as interactions are different at these two interfaces. Nonequilibrium MD (NEMD) simulations are performed using the LAMMPS package[46] with the trained NNPs to determine the TBCs. More details of the NNP development and MD simulations are provided in the Supporting Information.

## 3. RESULTS AND DISCUSSION

TDTR measurements were performed on six samples with varying AlN thicknesses at 300 and 500 K. Figure 2 illustrates

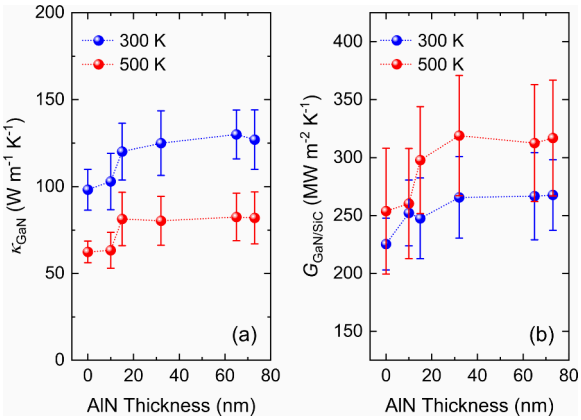



**Figure 2.** (a) Measured thermal conductivity of the GaN layers as a function of the AlN thickness at 300 and 500 K. (b) GaN/SiC TBC versus AlN thickness at 300 and 500 K. The error bars are calculated by a Monte Carlo method accounting for all possible error sources.

the variations of $\kappa_{GaN}$ and $G_{GaN/SiC}$ as functions of AlN thickness. The error bars, accounting for all potential error sources, are calculated using a Monte Carlo method.[15,42] Notably, our $\kappa_{GaN}$ value without a transition layer (~98 W m$^{-1}$ K$^{-1}$) agrees well with the reported value of MBE-grown GaN on SiC (~100 W m$^{-1}$ K$^{-1}$) at 300 K.[47] Introducing an AlN transition layer leads to an increase in $\kappa_{GaN}$, as demonstrated in Figure 2a. This can be attributed to the better crystalline quality of the GaN layer when the AlN transition layer is introduced. Additionally, we present the measured $G_{GaN/SiC}$ values as a function of AlN thickness at 300 and 500 K in Figure 2b. Clear enhancements in TBC are observed at both temperatures as the AlN transition layer thickness increases from 0 to 32 nm. Further increases in AlN thickness do not significantly affect TBC. Our measured $G_{GaN/SiC}$ values without an AlN layer are similar to the results from a previous study conducted at both 300 and 500 K (~225 and ~285 MW m$^{-2}$ K$^{-1}$, respectively).[7] Notably, the TBC is found to be higher at 500 K compared to 300 K, which is a commonly observed trend due to the increasing heat capacity as the temperature increases below the Debye temperatures (i.e., ~636 K for GaN[48] and ~1200 K for SiC[49]). While the transition layer already demonstrates an improvement in interfacial thermal transport at 300 K (~18% for the mean fitted TBC with a 32 nm-thick AlN layer), a more substantial mean enhancement is

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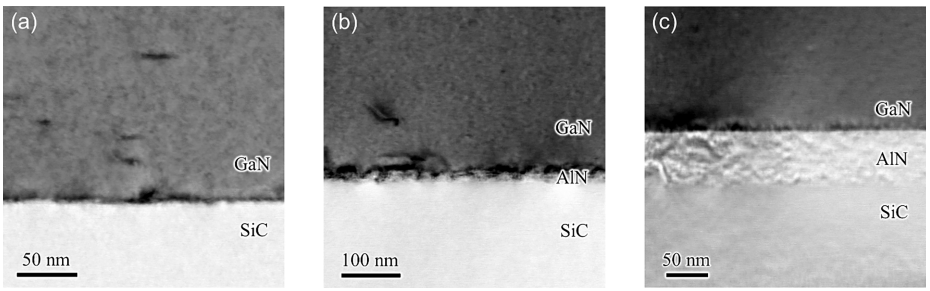



**Figure 3.** Bright-field STEM images of (a) sample I (no AlN), (b) sample III (15 nm-thick AlN), and (c) sample V (65 nm-thick AlN).

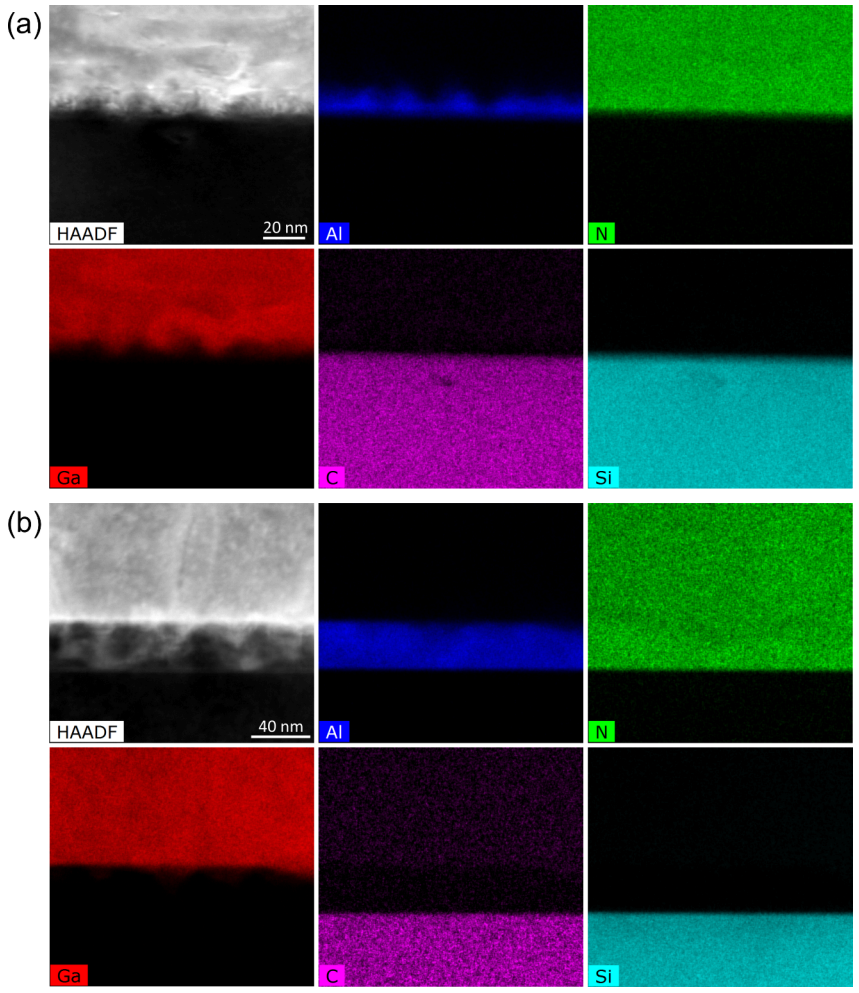



**Figure 4.** HAADF images and EDS maps of (a) sample II (10 nm-thick AlN) and (b) sample IV (32 nm-thick AlN). The Al, N, Ga, C, and Si maps are highlighted in blue, green, red, pink, and cyan, respectively.

observed at 500 K (~25% for the mean fitted TBC with a 32 nm-thick AlN layer). Furthermore, both the GaN effective resistance ($R_{GaN} = d_{GaN}/\kappa_{GaN}$, where $d_{GaN}$ is the GaN thickness) and the GaN/SiC resistance ($R_{GaN/SiC} = 1/G_{GaN/SiC}$) exhibit a comparable decreasing trend relative to AlN thickness, remaining relatively constant beyond 32 nm (see Figure S2 in the Supporting Information). This consistent behavior results in an overall reduction in the total resistance of the GaN layer and the GaN/SiC interface as the AlN thickness increases. These findings indicate that the incorporation of an AlN transition layer can indeed enhance thermal transport across GaN/SiC, particularly at elevated temperatures.

To investigate the relationship between thermal transport and the interface structure, we employed STEM and EDS to study the quality of the GaN/AlN/SiC interfaces. Figure 3 displays the bright-field STEM images of the interface structures for samples I, III, and V, respectively. In the case of the GaN/SiC interface without the AlN layer (sample I), lattice-mismatch-induced defects are observed both at the interface and within the GaN layer, represented by dark contrast features in Figure 3a. Upon introducing a 15 nm-thick AlN layer (sample III), the AlN layer is seen to consist of small islands at the interface (Figure 3b), which are associated with the three-dimensional growth mode of AlN on SiC commonly reported in the literature.[50,51] In contrast, sample V with a



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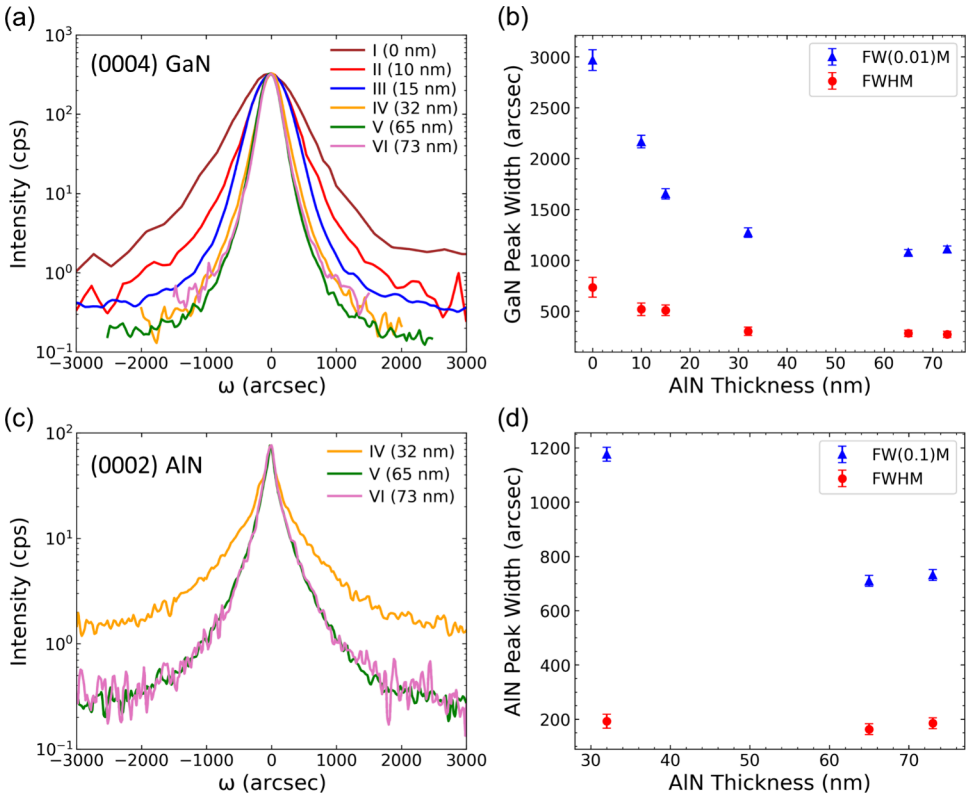



**Figure 5.** (a) Triple-axis rocking curves of (0004) GaN for all six samples. (b) (0004) GaN FWHM and FW(0.01)M rocking curve peak widths. (c) Triple-axis rocking curves of (0002) AlN for IV, V, and VI. (d) (0002) AlN FWHM and FW(0.1)M rocking curve peak widths. The scan peak intensities are normalized.

much thicker AlN layer exhibits no island features (Figure 3c), and a comparison of low-magnification STEM images reveals a reduction in dislocation density throughout the GaN layer with increasing AlN thickness (see Figure S3 in the Supporting Information). Furthermore, additional high-resolution STEM images and inverse fast Fourier transforms (IFFT) indicate a pristine AlN/SiC interface in sample V in contrast to the defects observed at the GaN/SiC interface in sample I (see Figure S4 in the Supporting Information).

In addition, high-angle annular dark-field (HAADF) images and EDS maps were obtained for two representative samples, II and IV, as shown in Figure 4. The EDS map confirms a chemically rough AlN/GaN interface for sample II, which has a 10 nm-thick AlN transition layer (Figure 4a), whereas a thicker AlN layer in sample IV results in smoother interfaces (Figure 4b). EDS maps for samples III and V can be found in the Supporting Information (Figures S5 and S6), which also demonstrates the disappearance of island morphology as the transition layer thickness increases. These characterization results indicate that the interface quality improves significantly with increasing AlN thickness (especially when AlN thickness >30 nm), which is a likely contributing factor to the enhanced TBC observed in the TDTR measurements.

TAXRD measurements were further conducted to evaluate the crystalline quality of GaN and AlN, as depicted in Figure 5. The amount of lattice mosaicity resulting from defects in the GaN layer depends on the structural characteristics of the underlying AlN transition layer, which exhibits a correlation with the enhancement of the AlN crystalline quality as the AlN thickness increases. Figure 5a illustrates the triple-axis rocking curves ($\omega$ scans) of the (0004) GaN for all six samples. Samples I, II, and III (with AlN thickness ≤15 nm) display significantly broader rocking curves, whereas narrower widths are observed for samples with thicker AlN layers. In the absence of AlN, the GaN peak widths for sample I are 700″ and 3000″ for the full width at half-maximum (FWHM) and full width at hundredth maximum (FW(0.01)M), respectively (Figure 5b). Here, FWXM ($X < 0.5$) widths are employed to quantify diffuse scattering intensity induced by defects. For sample IV, with 32 nm-thick AlN, the GaN FWHM and FW(0.01)M decrease to 300″ and 1280″, respectively. While the FWHM remains constant with thicker AlN layers, the FW(0.01)M further narrows to ∼1100″ for both samples with 65 and 73 nm-thick AlN (samples V and VI). This trend indicates that thicker AlN results in higher-quality GaN. In the case of GaN directly grown on SiC, the lattice mismatch (∼3.5%) leads to dislocations and defects in the GaN layer, which increase phonon scattering and consequently reduce the thermal conductivity.[47,S2] However, the lattice mismatch for AlN/SiC (∼1.0%) or GaN/AlN (∼2.5%) is smaller than that for GaN/SiC. The AlN serves as a buffer layer between GaN and SiC, reducing the dislocation density in GaN and thus improving its thermal conductivity, as evidenced by the increasing trend in GaN thermal conductivity shown in Figure 2a. The quality of AlN is also characterized by the triple-axis (0002) AlN rocking curves. As shown in Figure 5c,d, we observe greater diffuse scattering for the thinner AlN layer (sample IV) compared to the thicker layers (samples V and VI), indicating that the distortion in AlN decreases with an increased AlN thickness. The FW(0.1)M for 32 nm-thick AlN (sample IV) is 1180″, while it is 710″ for the 65 nm (sample V) and 730″ for the 73 nm (sample VI). Regarding the thermal

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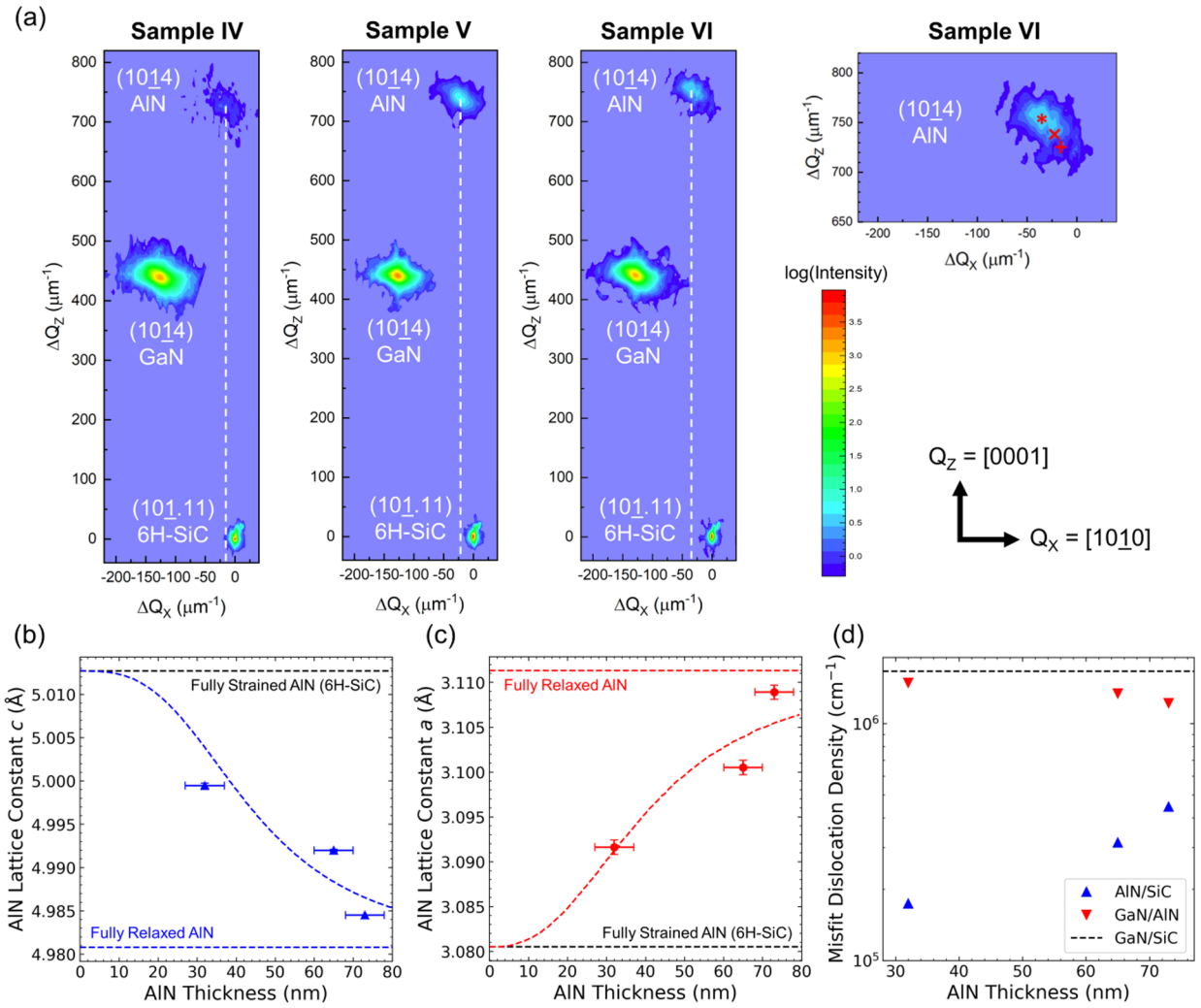



**Figure 6.** (a) Reciprocal space maps (RSMs) for determining lattice parameters of AlN in samples IV, V, and VI. The lattice parameters are determined by measuring the displacement of AlN peak from the 6H-SiC peak along $Q_X$ and $Q_Z$ directions. A magnified RSM for the sample VI (10$\bar{1}$4) AlN reciprocal lattice point is shown, whose peak is marked with a * symbol. The AlN positions for samples IV and V are marked with + and × symbols, respectively, on the same RSM to show the change in relaxation. (b) Out-of-plane lattice parameter $c$ and (c) in-plane lattice parameter $a$ of AlN in samples IV, V, and VI. (d) Calculated misfit dislocation density of both GaN/AlN and AlN/SiC interfaces based on the lattice mismatch.

conductivity of AlN, Aissa et al.[53] observed an increase from 80 to 175 W m$^{-1}$ K$^{-1}$ as the thickness of AlN films varied from 260 to 8000 nm. Given that our studies involve AlN films with lower thicknesses, we anticipate an even smaller thermal conductivity due to the crystalline quality of the film and the size effect of phonon transport. Overall, structures with fewer crystal imperfections could exhibit reduced phonon scattering, which facilitates the interfacial thermal transport. Therefore, we attribute the enhanced TBC in part to the improved crystal and interface quality resulting from the thicker AlN interlayers.

To characterize the structure of AlN, we measured the lattice parameters of AlN in samples IV, V, and VI, which have relatively thick AlN transition layers. Asymmetric (10$\bar{1}$4) GaN/AlN and (10$\bar{1}$.11) SiC reciprocal space measurements were performed to determine the in-plane lattice parameter (denoted as $a$) by measuring the displacement from the 6H-SiC peak along $Q_X$, as depicted in Figure 6a. Similarly, the out-of-plane lattice parameter $c$ can be measured along the out-of-plane direction $Q_Z$. In the reciprocal space maps (RSMs) shown in Figure 6a, while the AlN peak undergoes changes, the lattice parameters of the GaN layer remain relatively stable (almost the fixed position of the peak). We observe that both the (0002) and (10$\bar{1}$4) AlN peaks shift with varying AlN thicknesses (Figure 6a and Figure S7), indicating increasing relaxation with a thicker AlN layer. Figure 6b,c shows the lattice parameters $a$ and $c$ for AlN, and the uncertainties of the AlN thickness are depicted as well. It should be noted that for the bulk lattice parameters, $a_{6H\text{-}SiC}$ (3.08049 Å) < $a_{AlN}$ (3.11131 Å), which implies that we should observe out-of-plane expansion when the AlN layer is subjected to biaxial stress from the 6H-SiC substrate, as confirmed by Figure 6a. Based on the observed trend, the AlN lattice parameters $a$ and $c$ for samples II and III will lie between the fully strained AlN horizontal dotted lines (black) and sample IV (32 nm-thick AlN). The strain relaxation in the AlN layer does not have a detrimental impact on the TBC (Figure 2b). Regardless of the slight changes in the state of AlN, the GaN layer remains fully relaxed. At the AlN/SiC interface, the misfit dislocation density is estimated to be 1.8 × 10$^5$ cm$^{-1}$ (sample IV), 3.1 × 10$^5$ cm$^{-1}$ (sample V), and 4.5 × 10$^5$ cm$^{-1}$ (sample VI) based on the

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measured lattice parameters, as shown in Figure 6d. The expected misfit density at the AlN/GaN interface decreases slightly, indicating that the overall misfit density (the sum of the AlN/SiC and AlN/GaN interface levels) remains consistent among the samples. Hence, these results demonstrate that the fully relaxed GaN layers, together with the relatively low misfit densities at the interfaces, lead to consistent TBC for AlN thickness values ranging from 32 to 73 nm (Figure 2b).

It is noted that some previous experimental studies[9,11] have reported a different trend of temperature-dependent TBC for GaN/AlN/SiC structures, where the TBC decreases at higher temperatures. In these studies, phonon scattering resulting from near-interfacial defects and point defects within AlN was proposed as the cause of thermal resistance through the AlN layer. However, in our work, we achieved high-quality interfaces as the AlN thickness increases. Our structural characterizations confirm the improved crystalline quality and reduced imperfections near the interfaces. Although there is observable convergence in the TBC within the AlN thickness range of 32 to 73 nm, it remains uncertain whether this trend persists as the AlN thickness increases further. For thicker AlN (e.g., >100 nm), it becomes imperative to distinguish and analyze the thermal resistance of the AlN layer and the boundary resistances at the AlN interfaces separately. In such cases, the thermal resistance of AlN depends on both its thickness and its intrinsic thermal conductivity. Notably, as AlN reaches thicknesses on the order of a few microns, its thermal conductivity tends to converge toward the bulk value[54,55] and then AlN effective resistance increases with thickness. Eventually, when the AlN interlayer is too thick, its thermal resistance will be so significant that the thermal transport between SiC and GaN will be greatly impeded greatly.

At a higher temperature, such as 500 K, we experimentally observe an ∼25% mean enhancement in TBC when comparing samples with thick AlN to those without AlN. At 300 K, this enhancement is ∼18%. In prior investigations by Lee and Luo,[12] it was demonstrated that AlN demonstrates a favorable overlap with SiC and GaN in the vibrational spectrum. This spectral alignment plays a significant role in facilitating interfacial thermal transport through the elastic processes. This previous simulation, however, utilized the empirical Tersoff potential and exclusively predicted TBC at 300 K. To confirm this phonon-bridging effect, we calculated the TBCs using NEMD simulations with the trained NNPs discussed in Section 2 (see the Supporting Information for more details). The NNP-MD scheme has been demonstrated to accurately simulate semiconductor interfaces and predict thermal properties, comparable to first-principles calculations.[45,56,57] As shown in Table 1, we observe similar trends compared with our TDTR measurements, where the presence of the AlN transition layer enhances the TBC and the enhancement becomes larger at elevated temperatures. It is important to note that while NNP-MD-predicted TBCs are close to the reported simulation results using empirical potentials,[12,14] they are higher than our measured values (Figure 2b), which can be attributed to the assumption of a perfect lattice match and the absence of defects in MD simulations.

Table 1. NNP-MD-Predicted TBCs for GaN/SiC and GaN/AlN/SiC Interfaces at 300 and 500 K

| | thermal boundary conductance (MW m$^{-2}$ K$^{-1}$) | |
|---|---|---|
| temperature (K) | GaN/SiC | GaN/AlN/SiC |
| 300 | 529 | 546 |
| 500 | 615 | 657 |

Beyond the enhanced elastic contributions facilitated by vibrational spectrum overlap (see Figure S9 in the Supporting Information), the bridging effects induced by AlN may also boost inelastic phonon transport, facilitating interfrequency phonon energy communication and thereby benefiting TBC.[58] This anharmonic effect is known to intensify with rising temperatures and likely contributes to the heightened TBC enhancement observed after the insertion of an AlN layer at higher temperatures. The anharmonic effects on TBC have been substantiated in prior theoretical studies.[58−60] Overall, our MD results stand as computational evidence supporting enhanced thermal transport across GaN/SiC interfaces in the presence of AlN transition layers. Importantly, as the interfaces in our simulations are perfect in crystal structures and devoid of defects, we can confirm the phonon-bridging effects while mitigating the influence of the crystalline quality, which is inevitable in experiments. The fact that the AlN intermediate layer enhances TBC at such perfect interfaces suggests it indeed plays a "phonon bridge" role in interfacial thermal transport.

## 4. CONCLUSIONS

In summary, our experimental investigation underscores the significant impact of the AlN transition layer on the TBC of the GaN/SiC interfaces. Our work stands out by providing a direct and controlled comparison of TBCs for GaN/SiC interfaces with and without AlN under the same growth conditions. By systematically varying the AlN thickness (ranging from 0 to 73 nm), the study showcases a clear enhancement in TBC with increasing AlN thickness, highlighting the effectiveness of the AlN transition layer in facilitating thermal transport across GaN/SiC interfaces. Structural characterizations corroborate the improvements in the interface and crystalline quality with thicker AlN layers. These improvements led to the growth of high-quality GaN and AlN, effectively reducing phonon-defect scattering. Moreover, our MD simulations using deep learning-based potentials show that AlN enhances TBC for atomically perfect interfaces, with a more pronounced effect at elevated temperatures, suggesting its role as a phonon bridge for improved interfacial thermal transport.

## ■ ASSOCIATED CONTENT

### Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsami.3c16905.

TDTR data fitting; characterization of the interfaces with STEM, EDS, and triple-axis XRD; development of neural network potentials; and details of nonequilibrium MD simulations (PDF)

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### Notes

The authors declare no competing financial interest.

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